

ASU Macro Theory Qualifier

Module 2 Companion: OLG Theory, Equilibrium, and Intuition

Standalone Companion to the Module 2 OLG Practice Packet

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Orientation

The main Module 2 packet is deliberately practice-forward: it gets you quickly to templates and exam mutations. This companion is the conceptual bridge. Its purpose is to make the baseline overlapping-generations model feel natural enough that extensions no longer look like unrelated tricks.

The organizing sequence is:

Kaldor facts and Solow $\rightarrow$ exchange OLG $\rightarrow$ government and debt $\rightarrow$ production OLG $\rightarrow$ pensions, efficiency, het

The recurring exam skill is not memorizing a formula. It is seeing which lifetime budget, market-clearing condition, and normalization are implied by the environment.

How to use this companion

Read Sections 5–7 before doing endowment OLG exercises. Read Sections 8–9 before government/debt questions. Read Sections 10–13 before capital and Social Security questions. Use Section 6 whenever the algebra of growth rates, ratios, and normalizations feels slippery.

1 Source Inventory and Design Brief

1.1 Governing documents

Source	Type	Role in this companion	Extraction notes
/mnt/data/SKILL.md	PhD Lecture Notes Builder	Required inventory, blueprint, canonical models, derivation templates, exercises, answer sketches, and quality audit.	Markdown read.
/mnt/data/MACRO_MANIFEST.md	ASU Macro Manifest	Module 2 OLG scope, source hierarchy, notation, required derivations, and quality gates.	Markdown read.
/mnt/data/ASU_Macro_Module_2_OLG.tex	Existing Module 2 packet	Used as the companion target: this file emphasizes theory and intuition rather than duplicating the practice packet.	TeX read.

1.2 Primary OLG lecture and handout sources

Source	Type	Topics used	Extraction notes
/mnt/data/OG1.pdf	Handout: Overlapping Generations I	Exchange OLG timing, consumption allocation, feasibility, individual lending/borrowing problem, lifetime budget, intertemporal FOC, sequential market equilibrium, loan-market clearing, Pareto optimality, government, transfers, debt. Pages 34–53 are the core.	PDF has no parseable embedded text; page images were used.
/mnt/data/OG2.pdf	Handout: Overlapping Generations II	Production OLG primitives, demographics, endowments, technology, household problem, CRRA/log saving, equilibrium law of motion, competitive equilibrium definition, BGP facts, transition mechanics, golden rule, dynamic efficiency, PAYG pensions. Pages 65–84 are the core.	PDF has no parseable embedded text; page images were used.
/mnt/data/slides--classofSeptember16-2.pdf	Lecture slides	Government in an OG economy, tax sequences, government budget balance, feasibility with government consumption, intergenerational transfers, public debt, bond market clearing, debt-financed transfer exercise.	Text extracted.

Source	Type	Topics used	Extraction notes
/mnt/data/slides--classofSeptember19.pdf	Lecture slides	Ricardian equivalence, present-value tax liabilities, production with endogenous labor supply, intratemporal and intertemporal FOCs, growth example with $R = (1+g)/\beta$.	Text extracted.
/mnt/data/hw3-gustavo.pdf	Current OLG homework	Two-period endowment OLG, time-varying discount factors, preference heterogeneity, three-period exchange OLG.	Text available through uploaded-file parser.
/mnt/data/hw4-gustavo.pdf	Current OLG/government homework	Government consumption, labor supply, PAYG transfers with population growth, two-type growing endowment OLG, public pension variants.	Text available through uploaded-file parser.
/mnt/data/hw5-gustavo.pdf	Current OLG capital homework	Capital OLG laws, balanced growth, foreign aid transition, old labor supply, government consumption, TFP and capital-output ratios.	Text available through uploaded-file parser.

1.3 Past ASU macro qualifying-exam anchors

Year	Local file	OLG pattern	Relevance
2016	2016-06_*.tex	Two-period capital OLG with skilled/unskilled labor aggregator, steady-state return, skill-premium claim.	High
2017	2017-06_*.tex	OLG with labor-income taxes financing public infrastructure; household FOCs, government budget, steady-state private capital, tax claim.	High
2018	2018-06_*.tex	Three-period OLG, survival/death probability, life-cycle consumption and saving, aggregate savings, capital-labor dynamics.	High
2019	2019-06_*.tex	Two-period OLG with college/non-college old-age labor income; borrowing, capital law, dynamic-efficiency claim.	High
2020	2020-06_*.tex	Two-period OLG with PAYG Social Security, balanced pension formula, contribution-rate comparative static.	High

Year	Local file	OLG pattern	Relevance
2021	2021-06_*.tex	Two-person household participation threshold, secondary earner efficiency, aggregate labor, steady-state output and return claims.	High
2022	2022-06_*.tex	Home production, home-good taste heterogeneity, market hours, capital-labor law, GDP and return claims.	High
2023	2023-06_*.tex	Endowment OLG with two types, interest-rate clearing, no-trade case, income/loss correlation.	High
2024	2024-06_*.tex	OLG with market good, non-tradable home good, BGP with market hours, permanent home-productivity shock, return dynamics.	High

1.4 Design choices for this companion

This companion makes four deliberate choices.

1. It starts from economic history and growth facts, because OLG is easiest to understand as a response to what representative-agent Solow-style models can and cannot explain.
2. It gives formal sequential competitive equilibrium definitions repeatedly, because the handouts emphasize equilibrium objects and the exams frequently ask for definitions.
3. It pauses on ratios and normalizations, because most algebra mistakes in OLG come from mixing K_t/L_t , $K_t/(A_t L_t)$, aggregate variables, and per-cohort variables.
4. It uses a sample of handout-style exercises as worked applications, but does not try to reproduce every exercise. The goal is to make the existing exercise packet easier to follow.

2 Detailed Blueprint

Component	Plan
Objective	Build conceptual and algebraic fluency in OLG models: environments, optimization, sequential competitive equilibrium, stationary/BGP objects, transition dynamics, golden rule, public pensions, heterogeneity, and comparative statics.
Prerequisites	Static Lagrangians, log and CRRA utility, gross returns, Cobb-Douglas factor prices, Solow normalized variables, market clearing.
Core models	Exchange OLG; exchange OLG with government; public debt and Ricardian equivalence; production OLG with capital; PAYG pensions; heterogeneity by type, age, labor profile, home productivity, and participation cost.

Component	Plan
Main derivations	Lifetime budget; log and CRRA saving rules; sequential equilibrium; loan-market clearing and goods-market feasibility; capital law; stationary/BGP definitions; Cobb-Douglas steady state; transition map; golden rule; dynamic efficiency; PAYG law; heterogeneity aggregation.
Graphical intuition	Intertemporal budget line and endowment point; loan market excess saving; capital law $k_{t+1} = \psi(k_t)$; golden-rule consumption curve.
Exercise strategy	Exercises are grouped by setup, equilibrium definition, closed forms, transition/comparative statics, and Economist X claims. Each answer sketch includes the setup, key equation, result, intuition, and common mistake.
Estimated study use	Two 3-hour blocks for theory and derivations, plus one 2-hour exercise block.

3 Module Map

Topic	Why it matters	Source/exam anchor	Core equation	Trap
Kaldor facts and Solow	Motivates normalized variables and BGP language.	Solow background; OG2 BGP remarks.	$k = K/(AL)$	Confusing level growth with steady state.
Exchange OLG	Purest model of intergenerational trade.	OG1; 2023 qual; hw3.	$\sum_h s_t^h = 0$	Thinking no trade means price is not pinned down.
Lifetime budget	The universal setup step.	OG1; all OLG quals.	$c_1 + c_2/R = e_1 + e_2/R$	Putting old income in the wrong period.
Sequential CE	A repeated exam command.	OG1 slides.	Optimize, clear markets.	Listing allocations but no prices or clearing.
Government	Taxes change lifetime wealth; debt changes timing.	OG1; Sept. 16 slides; hw4.	$C_t + G_t = Y_t$	Forgetting old taxes/-transfers.
Ricardian equivalence	Clarifies when debt is neutral.	Sept. 19 slides.	Same PV taxes.	Applying it to intergenerational redistribution.

Topic	Why it matters	Source/exam anchor	Core equation	Trap
Production OLG	Backbone of capital questions.	OG2; 2016–2024 quals.	$K_{t+1} = L_t s_t$	Wrong denominator for k_{t+1} .
Stationary/BGP objects	Converts growth models into steady objects.	OG2; hw5.	$\tilde{c} = c/A$	Calling growing wages non-stationary.
Transition dynamics	Determines jumps versus gradual adjustment.	OG2 mechanics; 2024 qual.	K_t predetermined.	Making K_t jump.
Golden rule	Efficiency benchmark.	OG2; 2019 qual.	$R^{GR} = (1+n)(1+g)$	Comparing net to gross returns.
PAYG pensions	Classic policy OLG application.	OG2; 2020 qual; hw4.	$B = (1+n)T$	Missing the pension return.
Heterogeneity	Most qualifier mutations are type heterogeneity.	2016, 2019, 2021–2024.	$K' = \sum_i N_i s_i$	Averaging prices instead of quantities.

4 Learning Objectives

After studying these notes, you should be able to:

1. Explain why OLG models are natural descendants of Solow-style growth analysis but add age, finite lives, and intergenerational trade.
2. Write the exchange OLG household problem, solve it graphically and algebraically, and define a sequential competitive equilibrium.
3. Derive and interpret the log saving rule with old-age income, taxes, pensions, and labor income in both periods.
4. Prove why loan-market clearing is equivalent to goods-market feasibility in the baseline exchange economy.
5. State a government/debt equilibrium and explain the exact scope and limits of Ricardian equivalence.
6. Formulate the production OLG environment with demographics, endowments, technology, firm behavior, household optimization, and market clearing.
7. Derive the implicit and explicit capital laws and distinguish a steady state from a balanced growth path.
8. Use capital maps to describe transition dynamics after preference, productivity, tax, pension, home-productivity, and demographic shocks.
9. Derive the golden-rule condition and use it to diagnose dynamic efficiency or inefficiency.

10. Handle heterogeneity by aggregating type-specific savings and labor supply before normalizing.

5 From Kaldor Facts and Solow to OLG

5.1 Why begin with Kaldor facts?

Kaldor-style growth facts are empirical regularities that motivated modern growth theory: output per worker grows over long periods; capital per worker grows; the capital-output ratio is roughly stable; the return on capital is not trending without bound; factor income shares are roughly stable; and large cross-country income differences persist. These facts are not a list to memorize for OLG. They tell you which variables should be stationary in a growth model.

A model with labor-augmenting technological progress usually writes aggregate production as

$$Y_t = F(K_t, A_t L_t) = A_t L_t f(k_t), \quad k_t \equiv \frac{K_t}{A_t L_t}.$$

A balanced growth path has constant k_t , constant factor shares, constant capital-output ratio, and growing levels:

$$K_t, Y_t, C_t \text{ grow at } (1+n)(1+g), \quad y_t = Y_t/L_t \text{ and } c_t = C_t/L_t \text{ grow at } (1+g).$$

The phrase “steady state” in an economy with $g > 0$ almost always means a steady state in normalized variables.

Language trap

If A_t grows, the economy is not literally stationary in levels. It can still be in a stationary equilibrium after normalization. On an exam, write “steady state in efficiency units” or “balanced growth path” when $g > 0$.

5.2 What Solow contributes

The Solow model teaches the logic of capital accumulation:

$$K_{t+1} - (1 - \delta)K_t = sY_t.$$

After dividing by effective labor, the capital law becomes

$$(1+n)(1+g)k_{t+1} = sf(k_t) + (1-\delta)k_t.$$

In Solow, saving is exogenous. The central contribution is not a theory of saving but a theory of how constant saving, population growth, depreciation, and technological growth determine a long-run capital intensity.

The golden-rule capital stock in Solow maximizes steady-state consumption per effective worker:

$$\max_k f(k) - [(1+n)(1+g) - (1-\delta)]k.$$

The first-order condition is

$$f'(k^{GR}) = \delta + (1+n)(1+g) - 1.$$

Equivalently, the net marginal product equals the growth rate of aggregate effective labor.

5.3 What Solow cannot do and why OLG enters

Solow cannot answer several questions that appear naturally in macro policy:

- Why do young people save and old people dissave?
- How does population growth affect the return provided by a PAYG pension system?
- Can competitive equilibria overaccumulate capital?
- How do transfers across generations affect interest rates and welfare?
- What happens when old agents also work, or when households differ in labor profiles, home productivity, or participation costs?

OLG answers these by replacing an infinitely-lived representative household with finitely-lived cohorts. The old at date t cannot trade with the young at date $t + 1$ except through assets, government policy, or the capital stock that the young carry into old age. That generational boundary is why debt, pensions, and dynamic inefficiency become real issues.

The key conceptual step

In a representative-agent growth model, the same infinitely-lived household owns all assets and internalizes all future dates. In OLG, the person saving today is young, while the person consuming asset returns tomorrow is old and then exits. Because generations are not all alive at the same time, intergenerational reallocation can change allocations and welfare even when all markets clear competitively.

6 Ratio and Timing Bootcamp

Most OLG algebra mistakes are ratio mistakes. This section isolates the mechanics.

6.1 Gross returns and dates

Use gross returns unless the problem explicitly asks for net rates:

$$R_{t+1} = 1 + r_{t+1}.$$

A young agent born at t saves s_t at date t and receives $R_{t+1}s_t$ at date $t + 1$. If production uses capital, the return paid at $t + 1$ is pinned down by the marginal product of capital at $t + 1$:

$$R_{t+1} = 1 - \delta + F_K(K_{t+1}, A_{t+1}L_{t+1}).$$

This timing is why the production OLG capital law is often implicit: the saving chosen at t depends on R_{t+1} , and R_{t+1} depends on K_{t+1} , which is itself generated by saving at t .

6.2 Aggregate, per worker, and per effective worker

Suppose the young cohort has size N_t and labor efficiency is A_t . If each young saves s_t , then

$$K_{t+1} = N_t s_t.$$

If the normalized capital stock is capital per effective young worker,

$$k_t = \frac{K_t}{A_t N_t},$$

then

$$k_{t+1} = \frac{K_{t+1}}{A_{t+1}N_{t+1}} = \frac{N_t s_t}{A_t(1+g)N_t(1+n)} = \frac{s_t/A_t}{(1+n)(1+g)}.$$

If $s_t = A_t \tilde{s}_t$, then

$$(1+n)(1+g)k_{t+1} = \tilde{s}_t.$$

Exam trap: current versus next denominator

Never write $k_{t+1} = s_t/A_t$ unless N and A are constant and you are explicit about the normalization. The asset created by saving at t is K_{t+1} , so the denominator is the effective labor available at $t+1$.

6.3 Cobb-Douglas factor prices in efficiency units

For

$$Y_t = K_t^\alpha (A_t L_t)^{1-\alpha} = A_t L_t k_t^\alpha,$$

where $k_t = K_t/(A_t L_t)$, the intensive production function is $f(k) = k^\alpha$. Competitive factor prices are

$$\begin{aligned} R_t &= 1 - \delta + f'(k_t) = 1 - \delta + \alpha k_t^{\alpha-1}, \\ w_t &= A_t [f(k_t) - k_t f'(k_t)] = A_t (1 - \alpha) k_t^\alpha. \end{aligned}$$

Here w_t is the wage per unit of raw labor when one unit of raw labor supplies A_t efficiency units. The wage per efficiency unit is

$$\tilde{w}(k_t) = f(k_t) - k_t f'(k_t) = (1 - \alpha) k_t^\alpha.$$

Three prices to keep separate

1. R_t : gross return paid to capital owners between $t-1$ and t .
2. w_t : wage per raw worker, typically grows with A_t on a BGP.
3. $\tilde{w}_t = w_t/A_t$: wage per efficiency unit, stationary on a BGP.

7 Baseline Exchange OLG

7.1 Environment and timing

Time is discrete, $t = 1, 2, \dots$. At each date a new generation is born. Individuals live for two periods: young at birth date t , old at date $t+1$. Generation 0 is the initial old. In the pure exchange model, there is one perishable good at each date, no storage, no capital, and no production unless endowments are interpreted as output.

Let N_t be the size of generation t . Agent h in generation t has endowments

$$\omega_t^h(t) \quad \text{when young}, \quad \omega_t^h(t+1) \quad \text{when old}.$$

Consumption is

$$c_t^h(t) \quad \text{when young}, \quad c_t^h(t+1) \quad \text{when old}.$$

The typical utility function is

$$u_t^h(c_t^h(t), c_t^h(t+1)),$$

strictly increasing, twice differentiable, and strictly concave. The log special case is

$$\ln c_t^h(t) + \beta_h \ln c_t^h(t+1).$$

7.2 Consumption allocations and feasibility

A consumption allocation is an assignment of consumption to every living agent at every date:

$$C = \left\{ (c_t^h(t), c_t^h(t+1))_{h=1}^{N_t} \right\}_{t \geq 1}.$$

At date t , the living agents are the young born at t and the old born at $t-1$. Feasibility requires

$$\sum_{h=1}^{N_t} c_t^h(t) + \sum_{h=1}^{N_{t-1}} c_t^h(t-1) \leq \sum_{h=1}^{N_t} \omega_t^h(t) + \sum_{h=1}^{N_{t-1}} \omega_{t-1}^h(t).$$

If there is no government consumption and goods are perishable, feasibility usually holds with equality in equilibrium because preferences are strictly increasing.

7.3 Sequential market arrangement

There is a one-period lending and borrowing market. A young agent can lend ℓ_t^h units of the date- t good and receive $R_t \ell_t^h$ units of the date- $(t+1)$ good when old. The sign convention is:

$$\ell_t^h > 0 \text{ means lending/saving,} \quad \ell_t^h < 0 \text{ means borrowing.}$$

The old cannot borrow from the future because they do not live to repay.

The young agent's sequential budget constraints are

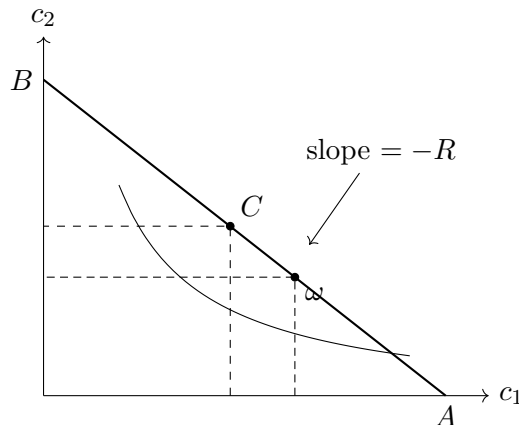
$$\begin{aligned} c_t^h(t) &\leq \omega_t^h(t) - \ell_t^h, \\ c_t^h(t+1) &\leq R_t \ell_t^h + \omega_{t+1}^h(t+1). \end{aligned}$$

Combining them gives the lifetime budget constraint

$$c_t^h(t) + \frac{c_t^h(t+1)}{R_t} \leq \omega_t^h(t) + \frac{\omega_{t+1}^h(t+1)}{R_t}. \quad (1)$$

The right side is lifetime wealth measured in date- t goods.

7.4 Graphical intuition: the intertemporal budget line



The endowment point $\omega = (\omega_1, \omega_2)$ lies on the budget line. If the chosen point C has $c_1 < \omega_1$, the agent is a lender. If $c_1 > \omega_1$, the agent is a borrower. The price R rotates the budget line around the endowment point: a higher R makes future consumption cheaper in terms of current consumption and steepens the budget line.

7.5 Optimality conditions

The Lagrangian for the lifetime problem is

$$\mathcal{L} = u(c_1, c_2) + \lambda \left(\omega_1 + \frac{\omega_2}{R} - c_1 - \frac{c_2}{R} \right).$$

Interior FOCs are

$$\begin{aligned} u_1(c_1, c_2) &= \lambda, \\ u_2(c_1, c_2) &= \frac{\lambda}{R}. \end{aligned}$$

Therefore

$$R = \frac{u_1(c_1, c_2)}{u_2(c_1, c_2)}. \quad (2)$$

The gross interest rate equals the marginal rate of substitution between current and future consumption.

If utility is separable, $u(c_1, c_2) = v(c_1) + \beta v(c_2)$, then

$$v'(c_1) = \beta R v'(c_2).$$

For log utility,

$$\frac{1}{c_1} = \beta R \frac{1}{c_2} \implies c_2 = \beta R c_1.$$

7.6 The log saving rule with old-age income

Let

$$e_1 = \omega_1, \quad e_2 = \omega_2,$$

and write the lifetime budget as

$$c_1 + \frac{c_2}{R} = e_1 + \frac{e_2}{R} \equiv W.$$

For log utility,

$$\begin{aligned} c_1 &= \frac{W}{1 + \beta}, \\ c_2 &= \frac{\beta R W}{1 + \beta}. \end{aligned}$$

Saving is $s = e_1 - c_1$, so

$$s = \frac{\beta}{1 + \beta} e_1 - \frac{1}{1 + \beta} \frac{e_2}{R}. \quad (3)$$

This is the single most useful OLG formula in the course.

Reading the formula

Young income raises saving because the agent wants to shift resources to old age. Old-age income lowers saving because it already finances old-age consumption. A higher return R lowers the present value of old-age income, which raises saving for given e_1, e_2 in the log case with old-age income. If $e_2 = 0$, log saving is independent of R .

7.7 Sequential competitive equilibrium without government

Sequential competitive equilibrium: exchange OLG

A sequential competitive equilibrium is a consumption allocation $\{c_t^h(t), c_t^h(t+1)\}$, a sequence of lending/borrowing decisions $\{\ell_t^h\}$, and a sequence of gross interest rates $\{R_t\}_{t \geq 1}$ such that:

1. Given $\{R_t\}$, each agent maximizes lifetime utility subject to the sequential budget constraints.
2. The loan market clears at every date:

$$\sum_{h=1}^{N_t} \ell_t^h = 0.$$

3. The initial old consume their endowments plus any assets specified as initial conditions.

Goods-market clearing need not be listed separately if the loan market clears and the initial old have no outstanding debt. To see the logic, aggregate young budget constraints:

$$\sum_h c_t^h(t) = \sum_h \omega_t^h(t) - \sum_h \ell_t^h.$$

Aggregate old budget constraints:

$$\sum_h c_{t-1}^h(t) = R_{t-1} \sum_h \ell_{t-1}^h + \sum_h \omega_{t-1}^h(t).$$

If loans cleared last period and clear this period, aggregate consumption equals aggregate endowment. Conversely, feasibility and old budget constraints imply loan-market clearing.

Why R is pinned down even if nobody trades

If all agents are identical, equilibrium has $\ell_t = 0$. But R_t is not arbitrary. It must make $\ell_t = 0$ optimal. The equilibrium price is the price that supports autarky.

7.8 Worked baseline: identical agents and no trade

Assume constant cohort size, identical agents, log utility, and endowments (ω_y, ω_o) . The saving rule is

$$s_t = \frac{\beta}{1+\beta} \omega_y - \frac{1}{1+\beta} \frac{\omega_o}{R_t}.$$

Loan-market clearing with identical agents requires $s_t = 0$. Hence

$$R_t = R = \frac{\omega_o}{\beta \omega_y}. \quad (4)$$

Equilibrium allocations are autarkic:

$$c_{1t} = \omega_y, \quad c_{2,t+1} = \omega_o.$$

For the handout example with $\beta = 1$ and $(\omega_y, \omega_o) = (3, 1)$,

$$R = \frac{1}{3}, \quad (c_y, c_o) = (3, 1).$$

Nobody saves or borrows, but the interest rate is low because agents are endowed with many goods when young and few when old. The price must make them willing not to shift resources across age.

7.9 Worked baseline: two endowment types and trade

Suppose there are two equal-mass types, both with log utility and $\beta = 1$:

$$(\omega_y^1, \omega_o^1) = (1, 1), \quad (\omega_y^2, \omega_o^2) = (2, 1).$$

Savings are

$$s^1 = \frac{1}{2} - \frac{1}{2R}, \quad s^2 = 1 - \frac{1}{2R}.$$

Loan clearing gives

$$0 = s^1 + s^2 = \frac{3}{2} - \frac{1}{R} \implies R = \frac{2}{3}.$$

Then type 1 borrows and type 2 lends:

$$s^1 = -\frac{1}{4}, \quad s^2 = \frac{1}{4}.$$

The interest rate reflects aggregate young resources relative to aggregate old resources. It is not an average of type-specific desired returns.

7.10 Preference heterogeneity and lender/borrower thresholds

Suppose all agents have equal endowments (ω, ω) but differ in discount factors β_h . Log saving is

$$s_h = \frac{\beta_h}{1 + \beta_h} \omega - \frac{1}{1 + \beta_h} \frac{\omega}{R} = \frac{\omega}{1 + \beta_h} \left(\beta_h - \frac{1}{R} \right).$$

Therefore

$$s_h > 0 \iff \beta_h > \frac{1}{R}.$$

There is a threshold discount factor

$$\hat{\beta} = \frac{1}{R}.$$

Agents more patient than $\hat{\beta}$ lend; agents less patient borrow. Loan clearing pins down R so that total desired lending equals total desired borrowing.

7.11 Endowment heterogeneity and the 2023 qualifier pattern

Let two types $i \in \{A, B\}$ have young endowment z_i and old endowment $z_i(1 - \delta_i)$, with equal masses. Log utility gives

$$s_i = \frac{\beta}{1 + \beta} z_i - \frac{1}{1 + \beta} \frac{z_i(1 - \delta_i)}{R}.$$

Loan clearing is

$$\sum_i s_i = 0 \implies R = \frac{\sum_i z_i(1 - \delta_i)}{\beta \sum_i z_i}.$$

Since $\delta_i \geq 0$, aggregate old endowment is below aggregate young endowment unless all $\delta_i = 0$, so $R < 1/\beta$.

If $\delta_A = \delta_B = \bar{\delta}$, then

$$R = \frac{1 - \bar{\delta}}{\beta},$$

and each type has zero saving:

$$s_i = \frac{z_i}{1 + \beta} \left[\beta - \frac{1 - \bar{\delta}}{R} \right] = 0.$$

There is no trade even though endowment levels differ, because both types have the same age profile.

If $z_A > z_B$, $\delta_A = \bar{\delta} + \Delta$, and $\delta_B = \bar{\delta} - \Delta$, then

$$\sum_i z_i(1 - \delta_i) = (1 - \bar{\delta})(z_A + z_B) - \Delta(z_A - z_B),$$

so the equilibrium interest rate is lower than in the common-loss case. The high-endowment young are the ones with larger old-age losses, so the economy has more aggregate desire to transfer resources to old age.

7.12 Pareto optimality in the exchange OLG

OLG economies can be competitive yet inefficient because an infinite chain of generations creates room for transfers from young to old at every date. Consider a stationary symmetric exchange economy with constant N , aggregate endowment Y , and allocation (c_1, c_2) for every generation. Feasibility requires

$$Nc_1 + Nc_2 = Y, \quad \text{or} \quad c_1 + c_2 = Y/N.$$

The handout's key result: in a stationary symmetric endowment economy with endowment (ω_1, ω_2) and strictly convex indifference curves, the autarkic competitive equilibrium is Pareto optimal if and only if the MRS at the endowment point is at least one:

$$\frac{u_1(\omega_1, \omega_2)}{u_2(\omega_1, \omega_2)} \geq 1.$$

If the MRS is less than one, the young value current goods relatively little compared with the old. A small transfer from every young cohort to every old cohort can make all future generations better off while also helping the initial old.

For the handout example $(\omega_y, \omega_o) = (3, 1)$ and $u = \log c_y + \log c_o$,

$$\text{MRS} = \frac{u_y}{u_o} = \frac{1/3}{1} = \frac{1}{3} < 1.$$

The allocation $(2, 2)$ is feasible and Pareto improves on $(3, 1)$ for all non-initial generations because

$$\log 2 + \log 2 > \log 3 + \log 1,$$

and it improves the initial old because they receive more consumption.

Why this is not a contradiction of the welfare theorem

The usual first welfare theorem has hidden assumptions about the commodity space and feasible reallocations. In OLG, there are infinitely many dated agents and the value of aggregate endowments can be finite in some price systems but not in others. Competitive equilibria need not be Pareto optimal in the same way representative-agent equilibria are.

8 Government in the Exchange OLG

8.1 Balanced government budget

Let $\tau_t^h(t)$ be taxes paid by generation t when young and $\tau_t^h(t+1)$ taxes paid when old. Negative taxes are transfers. Let G_t denote government consumption at date t . A balanced government budget is

$$G_t = \sum_{h=1}^{N_t} \tau_t^h(t) + \sum_{h=1}^{N_{t-1}} \tau_t^h(t+1). \quad (5)$$

Government consumption is not valued by households in the baseline handouts. Feasibility becomes

$$C_t + G_t = Y_t,$$

where C_t is aggregate private consumption by the young and old alive at t .

The lifetime budget constraint becomes

$$c_t^h(t) + \frac{c_t^h(t+1)}{R_t} \leq \omega_t^h(t) - \tau_t^h(t) + \frac{\omega_t^h(t+1) - \tau_t^h(t+1)}{R_t}. \quad (6)$$

The FOC is unchanged because taxes shift lifetime wealth, not the relative price, when taxes are lump sum.

Sequential competitive equilibrium: exchange OLG with balanced-budget government

Given a tax sequence $\mathcal{T} = \{\tau_t^h(t), \tau_t^h(t+1)\}$, a sequential competitive equilibrium is a government consumption sequence $\{G_t\}$, a consumption allocation, lending/borrowing decisions, and gross interest rates $\{R_t\}$ such that:

1. Given taxes and $\{R_t\}$, households maximize utility subject to their tax-adjusted lifetime budgets.
2. Loan markets clear: $\sum_{h=1}^{N_t} \ell_t^h = 0$ for all t .
3. The government budget balances: equation (5) holds for all t .
4. Feasibility holds: aggregate private consumption plus G_t equals aggregate endowment/output.

8.2 Intergenerational transfers: young tax, old benefit

Assume identical agents, constant cohort size N , log utility with $\beta = 1$, endowments (ω_y, ω_o) , a tax τ on each young, and a transfer b to each old. With no government consumption and constant cohort size, budget balance implies $b = \tau$.

The saving rule is

$$s = \frac{\omega_y - \tau}{2} - \frac{\omega_o + b}{2R}.$$

Loan clearing $s = 0$ and $b = \tau$ imply

$$R = \frac{\omega_o + \tau}{\omega_y - \tau}. \quad (7)$$

The transfer raises the equilibrium interest rate when it shifts goods from the young to the old. Why? Young after-tax resources fall, old-age resources rise, and the price must make the young willing to neither borrow nor lend.

If population grows at rate n , the same per-young tax τ finances per-old benefit

$$b_{t+1} = (1 + n)\tau,$$

because there are $(1 + n)$ young per old next period. The implicit gross return of the PAYG transfer is $(1 + n)$ in an endowment economy without wage growth.

8.3 Fluctuating government consumption

If only the young pay taxes and government consumption per young is $g_t = G_t/N$, then $\tau_t = g_t$. With log utility and identical agents,

$$R_t = \frac{\omega_o}{\omega_y - g_t}.$$

If $G_t = G_H$ in even periods and G_L in odd periods, with $G_H > G_L$, then

$$R_t = \frac{\omega_o}{\omega_y - G_H/N} \quad \text{when } t \text{ is even,} \quad R_t = \frac{\omega_o}{\omega_y - G_L/N} \quad \text{when } t \text{ is odd.}$$

High government consumption absorbs young resources, so young after-tax income is low and the equilibrium interest rate is high.

9 Public Debt and Ricardian Equivalence

9.1 Government bonds

Let the government issue one-period bonds. A unit of bond purchased at date t pays one unit of the good at date $t + 1$. Let p_t be the date- t price of a bond and B_t the number of bonds sold by the government. The government budget constraint is

$$\sum_{h=1}^{N_t} \tau_t^h(t) + \sum_{h=1}^{N_{t-1}} \tau_{t-1}^h(t) + p_t B_t = G_t + B_{t-1}. \quad (8)$$

The right side is current spending plus repayment of last period's bonds.

Let b_t^h be the quantity of government bonds purchased by young agent h . The young budget constraints are

$$\begin{aligned} c_t^h(t) &= \omega_t^h(t) - \tau_t^h(t) - \ell_t^h - p_t b_t^h, \\ c_t^h(t+1) &= \omega_t^h(t+1) - \tau_t^h(t+1) + R_t \ell_t^h + b_t^h. \end{aligned}$$

The lifetime budget is

$$c_t^h(t) + \frac{c_t^h(t+1)}{R_t} = \omega_t^h(t) - \tau_t^h(t) + \frac{\omega_t^h(t+1) - \tau_t^h(t+1)}{R_t} - b_t^h \left(p_t - \frac{1}{R_t} \right).$$

No arbitrage requires

$$p_t = \frac{1}{R_t}. \quad (9)$$

Otherwise agents would prefer only the asset with the higher return. When (9) holds, private loans and government bonds are perfect substitutes in household portfolios.

Define total saving into private loans and government bonds as

$$s_t^h \equiv \ell_t^h + p_t b_t^h.$$

The bond-market/loan-market clearing condition is

$$\sum_{h=1}^{N_t} s_t^h = p_t B_t = \frac{B_t}{R_t}. \quad (10)$$

There is no net private borrowing/lending after aggregating, but the young can collectively save by holding government debt.

Sequential competitive equilibrium with public debt

Given initial debt B_0 and initial old assets, an equilibrium with debt consists of taxes, government consumption, debt $\{B_t\}$, prices $\{R_t, p_t\}$, household allocations, private loans, and bond holdings such that households optimize, no arbitrage $p_t = 1/R_t$ holds, the government budget (8) holds, and aggregate saving clears the bond market as in (10).

9.2 Ricardian equivalence: what it says

Consider two tax policies τ and $\hat{\tau}$. If, for every individual alive in the same generation and for a given interest-rate path, the present value of lifetime taxes is the same,

$$\tau_t^h(t) + \frac{\tau_t^h(t+1)}{R_t} = \hat{\tau}_t^h(t) + \frac{\hat{\tau}_t^h(t+1)}{R_t}, \quad (11)$$

then the individual's lifetime budget set is unchanged. Therefore the individual's consumption choice is unchanged.

The handout/slide proposition can be stated as follows.

Ricardian equivalence in this OLG environment

Fix an initial tax/debt policy and its competitive equilibrium. Consider a new tax policy that preserves the present value of each individual's tax liability at the initial equilibrium interest rates. Then there exists a debt policy such that the original allocation and interest rates, together with the new taxes and new debt sequence, form a competitive equilibrium.

The proof has two parts. First, households still optimize because their lifetime budget sets are unchanged. Second, the government can choose a bond sequence to make the date-by-date government budgets and asset-market clearing conditions hold. This second part is an induction argument using the government's budget constraint.

9.3 What Ricardian equivalence does not say

Ricardian equivalence is not the claim that all debt is irrelevant. It is the claim that pure tax timing is irrelevant under strong conditions. It fails or becomes inapplicable when:

- the present value of taxes changes for some generation;
- debt shifts repayment to unborn generations that were not compensated;
- borrowing constraints bind;
- taxes are distortionary;
- private and government borrowing rates differ;
- households do not internalize descendants' taxes through operative bequests.

In finite-life OLG, a debt-financed transfer to the current old paid by future young is not Ricardian neutral. The current old receive a windfall but do not repay it. Future generations face different lifetime resources.

9.4 Debt-financed transfer exercise

Consider the slide exercise: identical agents, $\beta = 1$, endowments $(\omega_y, \omega_o) = (2, 1)$, no government consumption, and an unexpected transfer $\Delta = 1/2$ to current old at t_0 , financed by debt and repaid by the young at $t_0 + 1$.

Before the policy, the autarky-supporting interest rate is

$$R = \frac{\omega_o}{\omega_y} = \frac{1}{2}.$$

At t_0 , the young can buy government bonds issued to finance old transfers. The current old have extra resources. The interest rate rises to make the young willing to hold the debt. The handout result is

$$R(t) = \frac{1}{2} \quad \text{if } t < t_0 \text{ or } t > t_0 + 1, \quad R(t_0) = 1, \quad R(t_0 + 1) = \frac{2}{3}.$$

The sequence is not neutral because different generations' lifetime resources change.

10 Production OLG with Capital

10.1 Primitives and environment

The production OLG model adds capital and firms to the life-cycle saving problem.

Demographics.

Time is discrete. Agents live for two periods, young and old. The size of the young cohort is L_t , with $L_{t+1} = (1 + n)L_t$.

Endowments.

A young agent supplies one unit of productive time inelastically in the baseline model. The initial old own the initial capital stock K_0 .

Technology.

Output is produced by competitive firms:

$$Y_t = F(K_t, A_t L_t) = A_t L_t f(k_t), \quad k_t = \frac{K_t}{A_t L_t},$$

with $A_{t+1} = (1 + g)A_t$.

Capital.

Capital depreciates at rate $\delta \in [0, 1]$. Firms rent capital at rental rate $\mathcal{R}_t$ and labor at wage w_t .

Preferences.

Baseline preferences are

$$u(c_{1t}, c_{2,t+1}) = v(c_{1t}) + \beta v(c_{2,t+1}).$$

For CRRA,

$$v(c) = \frac{c^{1-\sigma} - 1}{1-\sigma}, \quad \sigma > 0, \sigma \neq 1,$$

and for log utility $\sigma = 1$, $v(c) = \ln c$.

10.2 Firm problem and prices

Competitive firms solve

$$\max_{K_t, L_t} F(K_t, A_t L_t) - \mathcal{R}_t K_t - w_t L_t.$$

Constant returns imply zero profits and factor prices

$$\begin{aligned} \mathcal{R}_t &= F_K(K_t, A_t L_t) = f'(k_t), \\ w_t &= F_L(K_t, A_t L_t) = A_t [f(k_t) - k_t f'(k_t)]. \end{aligned}$$

The gross return to household saving from t to $t + 1$ is

$$R_{t+1} = 1 - \delta + f'(k_{t+1}). \tag{12}$$

The $t + 1$ subscript is crucial: saving at t buys capital used at $t + 1$.

10.3 Household optimization

The young born at t solves

$$\begin{aligned} \max_{c_{1t}, c_{2,t+1}, s_t} \quad & v(c_{1t}) + \beta v(c_{2,t+1}) \\ \text{s.t.} \quad & c_{1t} + s_t \leq w_t, \\ & c_{2,t+1} \leq R_{t+1} s_t, \\ & c_{1t}, c_{2,t+1} \geq 0. \end{aligned}$$

The lifetime budget is

$$c_{1t} + \frac{c_{2,t+1}}{R_{t+1}} \leq w_t.$$

Interior optimality gives the Euler equation

$$v'(c_{1t}) = \beta R_{t+1} v'(c_{2,t+1}). \tag{13}$$

10.4 CRRA saving rate

For CRRA utility, $v'(c) = c^{-\sigma}$, so

$$c_{1t}^{-\sigma} = \beta R_{t+1} c_{2,t+1}^{-\sigma} \implies c_{2,t+1} = (\beta R_{t+1})^{1/\sigma} c_{1t}.$$

Using the lifetime budget,

$$c_{1t} \left[1 + \beta^{1/\sigma} R_{t+1}^{(1-\sigma)/\sigma} \right] = w_t.$$

Savings are $s_t = w_t - c_{1t}$, so

$$s_t = \mathcal{S}(R_{t+1}) w_t, \quad \mathcal{S}(R) = \frac{\beta^{1/\sigma} R^{(1-\sigma)/\sigma}}{1 + \beta^{1/\sigma} R^{(1-\sigma)/\sigma}}. \quad (14)$$

This is the saving-rate function in the production handout.

The sign of $\mathcal{S}'(R)$ depends on the elasticity of intertemporal substitution $1/\sigma$:

$$\mathcal{S}'(R) \begin{cases} > 0, & \sigma < 1, \\ = 0, & \sigma = 1, \\ < 0, & \sigma > 1. \end{cases}$$

When $\sigma < 1$, substitution effects dominate: a higher return encourages saving. When $\sigma > 1$, income effects dominate: a higher return allows the target old-age consumption with less saving. In the log case, the two effects exactly cancel.

10.5 Log utility as the workhorse

With $v(c) = \ln c$,

$$c_{2,t+1} = \beta R_{t+1} c_{1t}.$$

The lifetime budget gives

$$c_{1t} = \frac{w_t}{1 + \beta}, \quad c_{2,t+1} = \frac{\beta R_{t+1} w_t}{1 + \beta}, \quad s_t = \frac{\beta}{1 + \beta} w_t.$$

The log saving rule is independent of R_{t+1} because the substitution and income effects cancel. This is why many ASU and OLG capital laws become explicit under log utility.

10.6 Capital-market clearing and the law of motion

At the end of date t , aggregate saving by the young becomes next period capital:

$$K_{t+1} = L_t s_t.$$

Using $w_t = A_t \tilde{w}(k_t)$ and dividing by $A_{t+1} L_{t+1} = A_t (1 + g) L_t (1 + n)$ gives the general law

$$(1 + n)(1 + g)k_{t+1} = \mathcal{S}(R_{t+1}) \tilde{w}(k_t), \quad R_{t+1} = 1 - \delta + f'(k_{t+1}). \quad (15)$$

This equation is implicit whenever $\mathcal{S}$ depends on R_{t+1} .

For Cobb-Douglas $f(k) = k^\alpha$ and log utility,

$$\tilde{w}(k_t) = (1 - \alpha) k_t^\alpha,$$

so

$$(1+n)(1+g)k_{t+1} = \frac{\beta}{1+\beta}(1-\alpha)k_t^\alpha. \quad (16)$$

The steady-state normalized capital stock is

$$k^* = \left[\frac{\beta}{1+\beta} \frac{1-\alpha}{(1+n)(1+g)} \right]^{\frac{1}{1-\alpha}}. \quad (17)$$

The steady-state gross return is

$$R^* = 1 - \delta + \alpha(k^*)^{\alpha-1}.$$

10.7 Sequential competitive equilibrium with capital

Sequential competitive equilibrium: production OLG

Given initial capital K_0 and initial old asset holdings, a sequential competitive equilibrium is a sequence of household allocations $\{c_{1t}, c_{2,t+1}, s_t\}_{t \geq 0}$, firm allocations $\{K_t, L_t, Y_t\}_{t \geq 0}$, prices $\{w_t, R_t\}_{t \geq 0}$, and aggregate capital $\{K_{t+1}\}_{t \geq 0}$ such that:

1. Given prices, households solve their lifetime problems.
2. Given prices, firms maximize profits and factor prices equal marginal products.
3. Capital-market consistency holds: $K_{t+1} = L_t s_t$.
4. Labor-market clearing holds: firms hire the labor supplied by the young.
5. Goods feasibility holds:

$$C_t + K_{t+1} - (1-\delta)K_t = Y_t,$$

where $C_t = L_t c_{1t} + L_{t-1} c_{2t}$.

The production handout shows that if households optimize, firms price competitively, and $K_{t+1} = L_t s_t$, then aggregate feasibility follows from household budget constraints, firm zero profit, and the no-arbitrage return condition.

10.8 Stationary equilibrium and balanced growth path

A stationary equilibrium in normalized variables is a path such that

$$k_t = k^*, \quad R_t = R^*, \quad \frac{w_t}{A_t} = \tilde{w}^*, \quad \frac{c_{1t}}{A_t} = \tilde{c}_1^*, \quad \frac{c_{2,t+1}}{A_{t+1}} = \tilde{c}_2^*.$$

If $g > 0$, the levels $w_t, c_{1t}, c_{2,t+1}$ grow over time even though normalized variables are constant. Aggregate output and aggregate consumption grow at $(1+n)(1+g)$; per-capita output and consumption grow at $(1+g)$.

Steady state versus BGP

- If $n = g = 0$, a steady state means constant levels.
- If $g > 0$ or $n > 0$, a balanced growth path means constant normalized variables and growing aggregates.
- On a BGP, old and young individual consumption both grow at $(1+g)$ across cohorts,

while aggregate consumption grows at $(1+n)(1+g)$.

11 Graphical Mechanics and Transitional Dynamics

11.1 The capital map

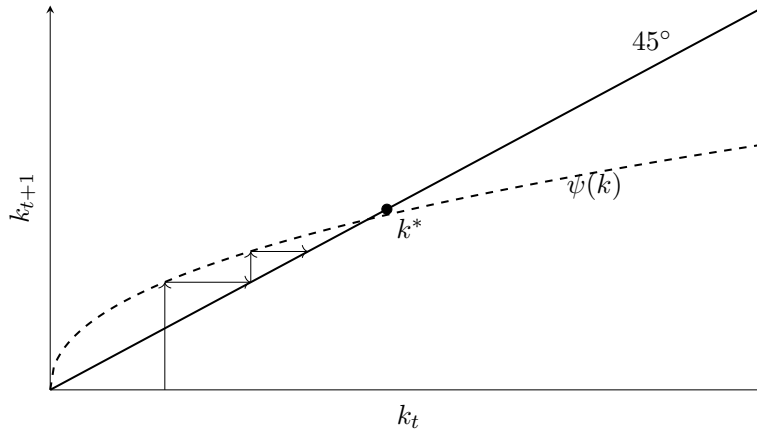
Write the capital law as

$$k_{t+1} = \psi(k_t).$$

In the log Cobb-Douglas case,

$$\psi(k) = \frac{\beta(1-\alpha)}{(1+\beta)(1+n)(1+g)} k^\alpha.$$

A steady state solves $k = \psi(k)$.



If ψ is increasing and crosses the 45-degree line once from above, convergence is monotone. If ψ is decreasing near the steady state, convergence can oscillate. If ψ is non-concave, there can be multiple steady states. In the log Cobb-Douglas case, ψ is increasing, concave, and has a unique positive steady state.

11.2 Predetermined capital

At date t_0 , K_{t_0} was chosen by the young at $t_0 - 1$. Therefore K_{t_0} cannot jump in response to a surprise at the start of t_0 . Prices can jump if the shock changes current productivity, current labor, or current effective labor. Future capital changes only through the saving decisions made at and after t_0 .

11.3 Permanent increase in patience

Suppose the economy is in steady state and β unexpectedly rises permanently. In the log Cobb-Douglas law,

$$\psi(k; \beta) = \frac{\beta(1-\alpha)}{(1+\beta)(1+n)(1+g)} k^\alpha.$$

A higher β raises $\beta/(1+\beta)$, shifting the map upward. At t_0 , k_{t_0} is predetermined. The young save more, so k_{t_0+1} rises. The economy converges monotonically to a higher k^* . The wage per efficiency unit rises and the gross return falls along the transition.

11.4 Permanent increase in TFP

Let production be $Y_t = ZK_t^\alpha(A_tL_t)^{1-\alpha}$. With log utility, wage income is proportional to Zk_t^α . A permanent increase from Z to Z' shifts the capital law up. At t_0 , K_{t_0} is predetermined, but wages and the rental rate can jump because current production is more productive. Capital rises gradually through higher saving.

A useful Cobb-Douglas property: in some log/full-depreciation cases, the steady-state return can be independent of the level of TFP. For example,

$$(1+n)(1+g)k' = \frac{\beta}{1+\beta}(1-\alpha)Zk^\alpha$$

implies

$$(k^*)^{1-\alpha} = \frac{\beta}{1+\beta} \frac{(1-\alpha)Z}{(1+n)(1+g)}.$$

Then

$$Z\alpha(k^*)^{\alpha-1} = \alpha \frac{(1+n)(1+g)(1+\beta)}{\beta(1-\alpha)},$$

which does not depend on Z . This is why the handout asks you to check whether the rental rate returns to its original steady-state level.

11.5 Temporary increase in TFP

If TFP rises only at t_0 and returns to normal at $t_0 + 1$, current wages and rental rates jump at t_0 . Higher current wages raise saving, so k_{t_0+1} rises. When TFP returns to normal, the capital stock is high relative to productivity, so the rental rate can fall below its original steady-state value before gradually returning.

Transition trap

For a temporary productivity shock, the return can overshoot in the opposite direction after the shock ends. The reason is that capital responds with a lag, while TFP jumps immediately.

11.6 Comparative statics checklist

For any shock, ask four questions in order:

1. Does the shock change current resources or preferences of the young at t ?
2. Does it change current production prices holding K_t fixed?
3. Does it change the saving map ψ and hence future K_{t+1} ?
4. Does it change the denominator used to define k_t (labor, effective labor, market hours, type composition)?

Most wrong comparative statics skip question 4.

12 Golden Rule and Dynamic Efficiency

12.1 Steady-state consumption in efficiency units

Aggregate feasibility is

$$C_t + K_{t+1} - (1-\delta)K_t = Y_t.$$

Divide by $A_t L_t$:

$$\frac{C_t}{A_t L_t} + (1+n)(1+g)k_{t+1} - (1-\delta)k_t = f(k_t).$$

On a BGP with $k_{t+1} = k_t = k$, total private consumption per effective worker is

$$\tilde{C}(k) = f(k) + (1-\delta)k - (1+n)(1+g)k = f(k) - [\delta + (1+n)(1+g) - 1]k. \quad (18)$$

12.2 Golden-rule capital stock

The golden-rule capital stock maximizes (18):

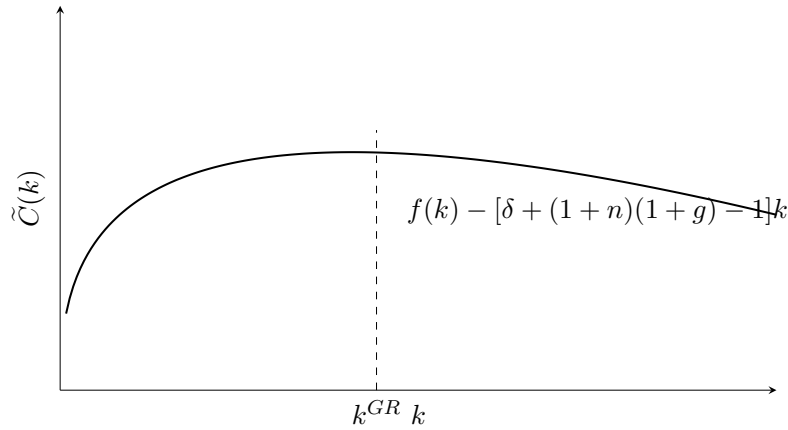
$$\max_k f(k) - [\delta + (1+n)(1+g) - 1]k.$$

The first-order condition is

$$f'(k^{GR}) = \delta + (1+n)(1+g) - 1. \quad (19)$$

Equivalently,

$$R^{GR} = 1 - \delta + f'(k^{GR}) = (1+n)(1+g). \quad (20)$$



12.3 Dynamic efficiency

A feasible allocation is dynamically efficient if there is no other feasible allocation, starting from the same initial capital, that gives at least as much aggregate consumption at every date and strictly more at some date.

In steady state, the diagnostic is:

$$\begin{cases} k^* > k^{GR} & \iff & R^* < (1+n)(1+g) & \text{dynamically inefficient,} \\ k^* \leq k^{GR} & \iff & R^* \geq (1+n)(1+g) & \text{dynamically efficient, in the steady-state test.} \end{cases}$$

Why? If $k^* > k^{GR}$, reducing capital slightly raises current consumption because less investment is needed, and it also raises future feasible consumption because the economy was on the downward side of the steady-state consumption curve.

12.4 Proof sketch for inefficiency when $k^* > k^{GR}$

At the original steady state,

$$\tilde{C}^* = f(k^*) + (1 - \delta)k^* - (1 + n)(1 + g)k^*.$$

Choose a small $\Delta < 0$ and set future capital to $k^* + \Delta$. At the transition date, investment falls by $-(1 + n)(1 + g)\Delta > 0$, so consumption can rise. In subsequent dates, the consumption change is approximately

$$\Delta \tilde{C} \approx [f'(k^*) + 1 - \delta - (1 + n)(1 + g)] \Delta = [R^* - (1 + n)(1 + g)] \Delta.$$

If $k^* > k^{GR}$, then $R^* < (1 + n)(1 + g)$. Since $\Delta < 0$, the product is positive. Consumption can be higher forever.

Gross versus net comparison

Do not compare r^* to $(1 + n)(1 + g)$. Compare gross $R^* = 1 + r^*$ to gross growth $(1 + n)(1 + g)$, or compare net r^* to $(1 + n)(1 + g) - 1$.

13 Public Pensions and PAYG Social Security

13.1 Simple PAYG system

Let young agents pay a tax T_t and old agents receive benefit B_{t+1} . With no labor-efficiency growth and population growth n , a balanced PAYG budget implies

$$L_t T_t = L_{t-1} B_t \implies B_{t+1} = (1 + n) T_t.$$

The implicit gross return on PAYG contributions is $(1 + n)$. With wage growth g , a proportional tax on wages gives an implicit return $(1 + n)(1 + g)$ along a BGP.

The young household solves

$$\begin{aligned} \max_{c_{1t}, c_{2,t+1}, s_t} \quad & v(c_{1t}) + \beta v(c_{2,t+1}) \\ \text{s.t.} \quad & c_{1t} + s_t = w_t - T_t, \\ & c_{2,t+1} = R_{t+1} s_t + B_{t+1}. \end{aligned}$$

The Euler equation remains

$$v'(c_{1t}) = \beta R_{t+1} v'(c_{2,t+1}).$$

The pension changes saving because it is old-age income.

For log utility,

$$s_t = \frac{\beta}{1 + \beta} (w_t - T_t) - \frac{1}{1 + \beta} \frac{B_{t+1}}{R_{t+1}}. \quad (21)$$

Substituting $B_{t+1} = (1 + n) T_t$ gives

$$s_t = \frac{\beta}{1 + \beta} w_t - \frac{T_t}{1 + \beta} \left[\beta + \frac{(1 + n)}{R_{t+1}} \right].$$

Social Security crowds out private saving. Whether total saving falls one-for-one with the tax depends on the comparison between the private return R and the PAYG return $(1 + n)$ (or $(1 + n)(1 + g)$ with wage growth).

13.2 Dynamic efficiency interpretation

If $R > (1+n)(1+g)$, private capital earns more than PAYG. A pension tax reduces saving, but households do not view PAYG as a perfect substitute for private saving because it has a lower return. If $R = (1+n)(1+g)$, PAYG and private capital have the same return at the margin. If $R < (1+n)(1+g)$, the economy is dynamically inefficient and PAYG can improve welfare by replacing low-return capital with a higher implicit intergenerational return.

The handout summarizes the saving response as:

$$\frac{\partial s}{\partial T} \begin{cases} < -1, & r < n & \text{dynamic inefficiency in net-return notation,} \\ = -1, & r = n & \text{golden rule,} \\ > -1, & r > n & \text{dynamic efficiency.} \end{cases}$$

In gross terms with wage growth, replace $1+r$ by R and $1+n$ by $(1+n)(1+g)$.

13.3 Proportional PAYG taxes: 2020 qualifier template

Suppose $d_t = \tau w_t$ and $\delta = 1$. The government budget balances period by period:

$$N_t^y d_t = N_t^o p_t = N_{t-1}^y p_t.$$

Since $N_t^y = (1+n)N_{t-1}^y$,

$$p_t = (1+n)\tau w_t. \quad (22)$$

A young agent born at t expects old-age pension

$$p_{t+1} = (1+n)\tau w_{t+1}.$$

With Cobb-Douglas and $k_t = K_t/L_t$ when there is no efficiency growth,

$$w_t = (1-\alpha)k_t^\alpha, \quad R_{t+1} = \alpha k_{t+1}^{\alpha-1} \quad (\delta = 1).$$

The log saving rule is

$$s_t = \frac{\beta}{1+\beta}(1-\tau)w_t - \frac{1}{1+\beta} \frac{(1+n)\tau w_{t+1}}{R_{t+1}}.$$

Since

$$\frac{w_{t+1}}{R_{t+1}} = \frac{(1-\alpha)k_{t+1}^\alpha}{\alpha k_{t+1}^{\alpha-1}} = \frac{1-\alpha}{\alpha} k_{t+1},$$

and $K_{t+1} = N_t^y s_t$, the normalized law is

$$(1+n)k_{t+1} = \frac{\beta}{1+\beta}(1-\tau)(1-\alpha)k_t^\alpha - \frac{1}{1+\beta}(1+n)\tau \frac{1-\alpha}{\alpha} k_{t+1}. \quad (23)$$

Solving for k_{t+1} ,

$$k_{t+1} = \frac{\frac{\beta}{1+\beta}(1-\tau)(1-\alpha)k_t^\alpha}{(1+n) \left[1 + \frac{\tau(1-\alpha)}{\alpha(1+\beta)} \right]}. \quad (24)$$

Thus a higher contribution rate lowers the capital-labor map through two channels: it reduces disposable young labor income and raises expected old-age pension income.

Why pensions reduce capital

In OLG, private capital is created by young saving. PAYG taxes lower resources available to save when young, and pensions reduce the need to save for old age. Both reduce K_{t+1} unless offset by other mechanisms.

14 Heterogeneity: How to Set It Up

The exams rarely ask only the representative young model. They mutate the baseline by adding types. The solution method is always:

solve each type's problem $\rightarrow$ aggregate quantities $\rightarrow$ normalize $\rightarrow$ interpret prices.

Do not average first-order conditions unless types are identical.

14.1 General type aggregation

Let type i have mass π_i in the young cohort, labor income e_{1t}^i , old-age income $e_{2,t+1}^i$, and log utility. Then

$$s_t^i = \frac{\beta}{1+\beta} e_{1t}^i - \frac{1}{1+\beta} \frac{e_{2,t+1}^i}{R_{t+1}}.$$

Aggregate saving is

$$K_{t+1} = N_t \sum_i \pi_i s_t^i.$$

Only after this step should you divide by the appropriate denominator.

14.2 Skilled and unskilled labor: 2016 pattern

Production is

$$F(K, S, U) = K^\alpha L^{1-\alpha}, \quad L = S^\phi U^{1-\phi}.$$

A fraction π of each cohort is skilled with z_s efficiency units; the rest are unskilled with z_u . Thus

$$S_t = \pi N_t z_s, \quad U_t = (1 - \pi) N_t z_u, \quad L_t = S_t^\phi U_t^{1-\phi}.$$

With log utility and no old-age labor income, aggregate saving is a constant fraction of aggregate labor income:

$$K_{t+1} = \frac{\beta}{1+\beta} (1 - \alpha) Y_t.$$

Therefore, in terms of $k_t = K_t/L_t$,

$$(1 + n)k_{t+1} = \frac{\beta}{1+\beta} (1 - \alpha) k_t^\alpha.$$

The steady-state k and return do not depend on z_s if the normalization uses the labor aggregator L_t .

The skill premium per worker is

$$\begin{aligned} \frac{\text{earnings of skilled worker}}{\text{earnings of unskilled worker}} &= \frac{F_S z_s}{F_U z_u} = \frac{\phi}{1 - \phi} \frac{U_t}{S_t} \frac{z_s}{z_u} \\ &= \frac{\phi}{1 - \phi} \frac{(1 - \pi) N_t z_u}{\pi N_t z_s} \frac{z_s}{z_u} = \frac{\phi(1 - \pi)}{(1 - \phi)\pi}. \end{aligned}$$

It is independent of z_s . The economist claim that raising z_s necessarily raises the skill premium is false in this production structure.

14.3 Old-age labor income: 2019 pattern

If agents work in both periods, old-age labor income enters $e_{2,t+1}$. For type i ,

$$e_{1t}^i = w_t a_y^i, \quad e_{2,t+1}^i = w_{t+1} a_o^i.$$

Saving is

$$s_t^i = \frac{\beta}{1+\beta} w_t a_y^i - \frac{1}{1+\beta} \frac{w_{t+1} a_o^i}{R_{t+1}}.$$

A type with a steep age-earnings profile, such as college agents with high old-age labor endowment, may borrow when young. The existence of borrowers is not a market failure. It is the equilibrium way to smooth consumption over a life cycle with rising earnings.

Old-age income trap

Do not use $s = \beta w / (1 + \beta)$ when old agents work or receive pensions. That formula assumes $e_2 = 0$.

14.4 Participation threshold: 2021 pattern

A two-person household has primary labor efficiency 1 and secondary efficiency $h \in (0, 1]$. If only the primary works, young labor income is w_t . If both work, income is $(1 + h)w_t$ and the household pays utility cost q .

For log utility and no old-age income, indirect utility from young income Y is

$$V(Y) = 2[(1 + \beta) \ln Y + \beta \ln(\beta R_{t+1}) - (1 + \beta) \ln(1 + \beta)].$$

Both work if

$$V((1 + h)w_t) - q \geq V(w_t).$$

The threshold is

$$q^* = 2(1 + \beta) \ln(1 + h). \quad (25)$$

It is time invariant because the wage cancels from the comparison. It is strictly increasing in h :

$$\frac{\partial q^*}{\partial h} = \frac{2(1 + \beta)}{1 + h} > 0.$$

The secondary-earner participation rate is $G(q^*)$ and aggregate labor per household is

$$\ell(h) = 1 + hG(q^*).$$

With Cobb-Douglas and log saving, the steady-state capital per unit of effective labor is independent of h , but total labor and total output can rise with h .

14.5 Home production: 2022 pattern

Agents value market consumption c and home goods h . Home goods are produced with time ℓ according to $h = G(\ell) = \ell^\lambda$, $\lambda \in (0, 1)$. Young agents split time between market work $m = 1 - \ell$ and home production. Old agents do not work in the market, so if home goods are valued and there is no leisure value, they devote all old-age time to home production.

For type γ , utility is

$$\ln c_{1t} + \gamma \ln h_{1t} + \beta [\ln c_{2,t+1} + \gamma \ln h_{2,t+1}].$$

The young problem implies

$$\frac{1}{c_{1t}} = \beta R_{t+1} \frac{1}{c_{2,t+1}}, \quad \frac{w_t}{c_{1t}} = \frac{\gamma \lambda}{\ell_t}.$$

Using $c_{1t} + s_t = w_t(1 - \ell_t)$ and $c_{2,t+1} = R_{t+1}s_t$, the saving share out of market earnings is still $\beta/(1 + \beta)$ and

$$\ell(\gamma) = \frac{\gamma \lambda}{1 + \beta + \gamma \lambda}, \quad m(\gamma) = 1 - \ell(\gamma) = \frac{1 + \beta}{1 + \beta + \gamma \lambda}. \quad (26)$$

Types with lower home-good taste γ work more in the market and save more in levels. But when the capital-labor ratio is normalized by market labor, the law can be independent of the type composition because aggregate saving is a constant fraction of aggregate market labor income.

14.6 Non-tradable home good and home-productivity shocks: 2024 pattern

The 2024-style problem has a market good c_1 and a non-storable, non-tradable home good $c_2 = Ah$. Preferences are

$$v(c_1, c_2) = \ln c_1 + \frac{c_2^{1-\sigma}}{1-\sigma}.$$

If the young splits time between market hours m_t and home hours $1 - m_t$, the young intratemporal FOC is

$$\frac{w_t}{c_{1t}} = A [A(1 - m_t)]^{-\sigma} = A^{1-\sigma} (1 - m_t)^{-\sigma}. \quad (27)$$

The intertemporal FOC is

$$\frac{1}{c_{1t}} = \beta R_{t+1} \frac{1}{c_{1,t+1}}.$$

With log utility over the market good and no old market income,

$$c_{1t} = \frac{w_t m_t}{1 + \beta}.$$

Substitute into (27):

$$\frac{1 + \beta}{m_t} = A^{1-\sigma} (1 - m_t)^{-\sigma}. \quad (28)$$

For fixed A , hours are constant over time. After a permanent change in A , hours generally change unless $\sigma = 1$. Therefore the claim that hours must remain constant merely because the economy was on a BGP before the shock is wrong.

If k_t is capital per effective unit of market labor, $k_t = K_t/(X_t N_t m_t)$, the capital law contains the ratio of current to next market hours:

$$(1 + n)(1 + g)k_{t+1} = \frac{\beta}{1 + \beta} (1 - \alpha) k_t^\alpha \frac{m_t}{m_{t+1}}.$$

After a permanent home-productivity shock, m_t jumps immediately. Since K_t is predetermined, the measured market capital-labor ratio and the return can jump on impact through the labor denominator even before new saving changes capital.

15 Public Infrastructure: Tax-Financed Productive Government Capital

The 2017 qualifier asks about a government that taxes young labor income to finance public infrastructure. A stylized version is:

$$y = z^\lambda k^\alpha \ell^{1-\alpha},$$

where z is public infrastructure available per worker. Young labor income is taxed at rate τ .

The household saving rule under log utility is

$$s_t = \frac{\beta}{1+\beta}(1-\tau)w_t.$$

Private capital accumulation is therefore weakened by taxes. But public infrastructure accumulation is strengthened by taxes. In a simple no-growth steady state with depreciation δ_z , the public law has the form

$$\delta_z z = \tau w(k, z),$$

while private capital satisfies

$$k = \frac{\beta}{1+\beta}(1-\tau)w(k, z).$$

The tax has two opposing effects:

- higher τ lowers disposable labor income and private saving;
- higher τ raises public infrastructure, which raises productivity and wages.

Therefore the claim “higher taxes finance infrastructure and therefore increase long-run private capital” is not generally valid. It requires parameter restrictions. The exam reward is to identify the opposing mechanisms, not to assert monotonicity.

16 Three-Period OLG

The 2018 qualifier uses three life stages: young, middle-aged, and old. The key is not a new idea but careful age indexing.

A generation born at t has utility

$$\ln c_t^y + \beta \ln c_{t+1}^m + \beta^2 \ln c_{t+2}^o.$$

Suppose labor income arrives only in middle age at wage w_{t+1} . The lifetime budget is

$$c_t^y + \frac{c_{t+1}^m}{R_{t+1}} + \frac{c_{t+2}^o}{R_{t+1}R_{t+2}} = \frac{w_{t+1}}{R_{t+1}}.$$

FOCs imply

$$\begin{aligned} \frac{1}{c_t^y} &= \beta R_{t+1} \frac{1}{c_{t+1}^m}, \\ \frac{1}{c_{t+1}^m} &= \beta R_{t+2} \frac{1}{c_{t+2}^o}. \end{aligned}$$

Thus

$$c_{t+1}^m = \beta R_{t+1} c_t^y, \quad c_{t+2}^o = \beta R_{t+2} c_{t+1}^m = \beta^2 R_{t+1} R_{t+2} c_t^y.$$

Substituting into the lifetime budget gives

$$(1 + \beta + \beta^2)c_t^y = \frac{w_{t+1}}{R_{t+1}}.$$

Therefore

$$\begin{aligned} c_t^y &= \frac{w_{t+1}}{R_{t+1}(1 + \beta + \beta^2)}, \\ c_{t+1}^m &= \frac{\beta w_{t+1}}{1 + \beta + \beta^2}, \\ c_{t+2}^o &= \frac{\beta^2 R_{t+2} w_{t+1}}{1 + \beta + \beta^2}. \end{aligned}$$

Savings by age are obtained from period-by-period asset accumulation. Young borrow to finance youth consumption, middle-aged save out of labor income, and old dissave. Aggregate savings at date t is the sum of young, middle-aged, and old savings alive at t . The capital law follows from $I_t = S_t$.

Three-period trap

Do not apply the two-period saving rule mechanically. First determine which age receives labor income and write the lifetime budget with the correct sequence of returns.

17 Core Derivation Templates

17.1 Template 1: Endowment OLG

Endowment OLG workflow

1. Write lifetime budget: $c_1 + c_2/R = e_1 + e_2/R$.
2. Derive FOC: $u_1/u_2 = R$ or $v'(c_1) = \beta R v'(c_2)$.
3. If log, write $s = \frac{\beta}{1+\beta} e_1 - \frac{1}{1+\beta} \frac{e_2}{R}$.
4. Aggregate: $\sum_i \pi_i s_i = 0$.
5. Solve for R or characterize it implicitly.
6. Interpret borrowers and lenders using the sign of s_i .

17.2 Template 2: Production OLG

Production OLG workflow

1. Define $k_t = K_t/(A_t L_t)$ or the ratio requested in the problem.
2. Write factor prices: $w_t = A_t[f(k_t) - k_t f'(k_t)]$, $R_{t+1} = 1 - \delta + f'(k_{t+1})$.
3. Solve household saving. For log/no old income, $s_t = \frac{\beta}{1+\beta} w_t$.
4. Aggregate: $K_{t+1} = L_t s_t$ or $K_{t+1} = \sum_i L_{it} s_{it}$.

5. Divide by next period's denominator.
6. Solve steady state or characterize $k_{t+1} = \psi(k_t)$.

17.3 Template 3: Government policy

Government/PAYG workflow

1. Write the government budget first. Determine whether it is balanced period by period or uses debt.
2. Put taxes in young resources and pensions/transfers in old resources.
3. Use the general log rule with $e_1 = w - T$ and $e_2 = B$ or old labor income.
4. Aggregate saving and normalize.
5. Interpret policy as changing disposable young resources, old-age resources, or productivity.

17.4 Template 4: Economist X claims

Evaluating claims

1. Translate the claim into a comparison of formal objects: k^* , R^* , Y/N , Y/L , hours, participation, or welfare.
2. Identify whether the shock changes a numerator, denominator, or both.
3. Derive the object being compared; do not rely on verbal intuition alone.
4. State whether the claim is always true, false, ambiguous, or true only under restrictions.
5. Give the economic mechanism in one sentence.

18 Closed-Form Catalog

Object	Formula	Use
Log saving with old income	$s = \frac{\beta}{1+\beta}e_1 - \frac{1}{1+\beta}\frac{e_2}{R}$	Endowment, pensions, old labor income
Identical endowment OLG return	$R = \omega_o/(\beta\omega_y)$	Supports autarky
Two-type endowment return	$R = \frac{\sum_i \pi_i e_2^i}{\beta \sum_i \pi_i e_1^i}$	2023 pattern
CRRA saving rate	$\mathcal{S}(R) = \frac{\beta^{1/\sigma} R^{(1-\sigma)/\sigma}}{1+\beta^{1/\sigma} R^{(1-\sigma)/\sigma}}$	Return-sensitive saving
Log Cobb-Douglas law	$(1+n)(1+g)k' = \frac{\beta}{1+\beta}(1-\alpha)k^\alpha$	Baseline production OLG

Object	Formula	Use
Log Cobb-Douglas steady state	$k^* = \left[\frac{\beta}{1+\beta} \frac{1-\alpha}{(1+n)(1+g)} \right]^{1/(1-\alpha)}$	Fast steady-state solution
Golden rule	$R^{GR} = (1+n)(1+g)$	Efficiency comparison
PAYG pension	$p_t = (1+n)\tau w_t$	2020 pattern
PAYG capital law	Equation (24)	Social Security comparative statics
Participation threshold	$q^* = 2(1+\beta) \ln(1+h)$	2021 pattern
Home time	$\ell(\gamma) = \frac{\gamma^\lambda}{1+\beta+\gamma^\lambda}$	2022 pattern
Skill premium	$\frac{\phi(1-\pi)}{(1-\phi)\pi}$	2016 pattern

19 Common Traps

High-frequency OLG mistakes

1. Using $s = \beta w / (1 + \beta)$ when the old have pensions, old-age labor income, or endowments.
2. Dividing K_{t+1} by $A_t L_t$ instead of $A_{t+1} L_{t+1}$.
3. Comparing r to $1 + n$ or R to n in dynamic-efficiency questions.
4. Saying an identical-agent endowment OLG has arbitrary R because no one trades.
5. Applying Ricardian equivalence to policies that change lifetime resources across generations.
6. Treating total output per person and output per market worker as the same object.
7. Forgetting that current capital is predetermined in transition questions.
8. Averaging type-specific prices instead of aggregating type-specific quantities.
9. Calling a growing economy nonstationary without checking normalized variables.
10. Assuming a tax-financed productive public input has a monotone effect on private capital.

20 Exercise Bank

20.1 Mechanical setup drills

- M1. Skill: lifetime budget.** A young agent has young income $w_t - T_t$, old labor income $a_o w_{t+1}$, and pension p_{t+1} . Write the lifetime budget and log saving rule.
- M2. Skill: equilibrium definition.** Define a sequential competitive equilibrium for the exchange OLG with a balanced-budget government and no debt.
- M3. Skill: capital law.** Suppose $K_{t+1} = N_t s_t$, $N_{t+1} = (1+n)N_t$, $A_{t+1} = (1+g)A_t$, and $s_t = A_t \tilde{s}(k_t)$. Derive the law for $k_{t+1} = K_{t+1} / (A_{t+1} N_{t+1})$.

M4. Skill: factor prices. For $Y = K^\alpha(AL)^{1-\alpha}$, derive w_t and R_t .

20.2 Closed-form/proof drills

- C1. Skill: endowment clearing.** In a two-type endowment OLG with masses π_i , young endowment z_i , old endowment $z_i(1 - \delta_i)$, and log utility, solve for R .
- C2. Skill: no-trade condition.** Show that if $\delta_i = \bar{\delta}$ for all types in C1, every type chooses zero saving.
- C3. Skill: CRRA saving.** Derive the CRRA saving rate $\mathcal{S}(R)$ and sign $\mathcal{S}'(R)$.
- C4. Skill: Cobb-Douglas steady state.** Derive (17) from (16).
- C5. Skill: golden rule.** Derive $R^{GR} = (1 + n)(1 + g)$ from the aggregate resource constraint.
- C6. Skill: PAYG law.** Derive equation (24) for proportional Social Security with $\delta = 1$.

20.3 Past-qual-style applications

- Q1. Skill: 2016 skilled labor.** In the skilled/unskilled model, show that the steady-state return is independent of z_s when capital is normalized by $L = S^\phi U^{1-\phi}$.
- Q2. Skill: 2021 participation.** Derive the participation threshold q^* and explain why it is increasing in h .
- Q3. Skill: 2022 home production.** With $G(\ell) = \ell^\lambda$, derive market hours for type γ and explain which type saves more.
- Q4. Skill: 2024 home productivity.** Use equation (28) to discuss whether market hours remain constant after a permanent increase in A .
- Q5. Skill: 2018 three-period OLG.** Derive the consumption profile in the three-period model where income arrives in middle age.

20.4 Trap and Economist X questions

- T1. Skill: no-trade price.** Economist X says that if all agents are identical in an endowment OLG, any interest rate clears the loan market. Is this correct?
- T2. Skill: Ricardian equivalence.** A government cuts taxes on the young today and taxes the same generation when old by an amount with the same present value. Is the allocation affected? What if repayment is shifted to the unborn?
- T3. Skill: dynamic efficiency.** An advisor says that a model with old-age labor income is automatically dynamically inefficient because young agents may borrow. Evaluate.
- T4. Skill: infrastructure tax.** Economist X says a higher infrastructure tax always increases long-run private capital. Evaluate.
- T5. Skill: TFP transition.** In a log Cobb-Douglas OLG, a one-period TFP increase occurs. Describe the qualitative path of the rental rate.

21 Answer Sketches

Mechanical setup drills

M1. Lifetime budget:

$$c_{1t} + \frac{c_{2,t+1}}{R_{t+1}} = w_t - T_t + \frac{a_o w_{t+1} + p_{t+1}}{R_{t+1}}.$$

Log saving:

$$s_t = \frac{\beta}{1+\beta}(w_t - T_t) - \frac{1}{1+\beta} \frac{a_o w_{t+1} + p_{t+1}}{R_{t+1}}.$$

Avoid dropping old income.

M2. Objects: tax sequence, government consumption, allocations, loans, interest rates. Conditions: household optimization, government budget balance, loan clearing, feasibility. The key new condition relative to the no-government economy is the government budget.

M3. Since $K_{t+1} = N_t s_t = N_t A_t \tilde{s}(k_t)$,

$$k_{t+1} = \frac{A_t N_t \tilde{s}(k_t)}{A_t(1+g)N_t(1+n)} = \frac{\tilde{s}(k_t)}{(1+n)(1+g)}.$$

The common mistake is dividing by $A_t N_t$.

M4. With $k = K/(AL)$,

$$Y = ALk^\alpha, \quad w = A(1-\alpha)k^\alpha, \quad R = 1 - \delta + \alpha k^{\alpha-1}.$$

Closed-form/proof drills

C1. Type saving is

$$s_i = \frac{\beta}{1+\beta} z_i - \frac{1}{1+\beta} \frac{z_i(1-\delta_i)}{R}.$$

Loan clearing gives

$$R = \frac{\sum_i \pi_i z_i (1-\delta_i)}{\beta \sum_i \pi_i z_i}.$$

C2. If $\delta_i = \bar{\delta}$, then $R = (1-\bar{\delta})/\beta$. Substitute into s_i to get $s_i = 0$ for every type. Different wealth levels do not create trade when age profiles are proportional.

C3. FOC gives $c_2 = (\beta R)^{1/\sigma} c_1$. Budget gives $c_1[1 + \beta^{1/\sigma} R^{(1-\sigma)/\sigma}] = w$. Hence

$$\mathcal{S}(R) = \frac{\beta^{1/\sigma} R^{(1-\sigma)/\sigma}}{1 + \beta^{1/\sigma} R^{(1-\sigma)/\sigma}}.$$

The sign follows from exponent $(1-\sigma)/\sigma$.

C4. In steady state,

$$(1+n)(1+g)k = \frac{\beta}{1+\beta}(1-\alpha)k^\alpha.$$

Divide by k^α and solve:

$$k^{1-\alpha} = \frac{\beta}{1+\beta} \frac{1-\alpha}{(1+n)(1+g)}.$$

C5. Steady-state consumption per effective worker is

$$\tilde{C}(k) = f(k) - [\delta + (1+n)(1+g) - 1]k.$$

FOC: $f'(k^{GR}) = \delta + (1+n)(1+g) - 1$. Add $1 - \delta$ to both sides to get $R^{GR} = (1+n)(1+g)$.

C6. Use $p_{t+1} = (1+n)\tau w_{t+1}$, $w_t = (1-\alpha)k_t^\alpha$, and $R_{t+1} = \alpha k_{t+1}^{\alpha-1}$. Substitute into the log saving rule and use $(1+n)k_{t+1} = s_t$ to obtain (23); solve for k_{t+1} to get (24).

Past-qual-style applications

Q1. Aggregate labor income is $(1-\alpha)Y_t$ regardless of how skilled efficiency z_s changes. With log utility, $K_{t+1} = \frac{\beta}{1+\beta}(1-\alpha)Y_t$. Since $Y_t = K_t^\alpha L_t^{1-\alpha}$, the normalized law in $k = K/L$ is independent of z_s . Thus R^* is unchanged.

Q2. Compare indirect utilities from income w and $(1+h)w$. The threshold is

$$q^* = 2(1+\beta)\ln(1+h),$$

so $\partial q^*/\partial h = 2(1+\beta)/(1+h) > 0$. Higher secondary productivity raises the consumption gain from joint work.

Q3. FOCs imply $s = \beta c_1$, $c_1 = w(1-\ell)/(1+\beta)$, and $(1+\beta)/(1-\ell) = \gamma\lambda/\ell$. Therefore

$$\ell(\gamma) = \frac{\gamma\lambda}{1+\beta+\gamma\lambda}, \quad m(\gamma) = \frac{1+\beta}{1+\beta+\gamma\lambda}.$$

Lower- γ types work more in the market and save more in levels.

Q4. Hours solve

$$\frac{1+\beta}{m} = A^{1-\sigma}(1-m)^{-\sigma}.$$

If A changes permanently, this equation generally implies a new m . The exception is $\sigma = 1$, where A drops out. Thus market hours need not remain constant after a home-productivity shock.

Q5. FOCs imply $c_m = \beta R_{t+1}c_y$ and $c_o = \beta R_{t+2}c_m$. Substitution into the lifetime budget yields

$$c_y = \frac{w_{t+1}}{R_{t+1}(1+\beta+\beta^2)}, \quad c_m = \frac{\beta w_{t+1}}{1+\beta+\beta^2}, \quad c_o = \frac{\beta^2 R_{t+2} w_{t+1}}{1+\beta+\beta^2}.$$

Trap and Economist X questions

T1. False. The loan market clears for any R only mechanically if everyone chooses zero saving, but zero saving is optimal only at $R = \omega_o/(\beta\omega_y)$ in the log identical-agent case. Prices must support optimality, not merely accounting.

T2. Same-generation tax timing with the same present value leaves lifetime budgets unchanged, so allocations can be unchanged with an appropriate debt policy. Shifting repayment to unborn generations changes lifetime resources across generations and is not neutral.

T3. False. Borrowing by young agents reflects life-cycle smoothing. Dynamic efficiency is about whether the steady-state return on capital is below the economy's growth rate. Check R^* versus $(1+n)(1+g)$.

T4. Ambiguous. A higher tax lowers disposable income and private saving but raises public infrastructure and productivity. The net effect on private capital depends on parameters.

T5. On impact, TFP raises the current marginal product of capital, so the rental rate jumps up. Higher saving raises next period capital. When TFP returns to normal, capital can be above its old steady state, so the rental rate can fall below its original level before converging back.

22 Past-Qual Practice Map

Year	Problem theme		What to practice in this companion	Sections
2016	Skilled/unskilled production OLG	pro-	Labor aggregator, type earnings, normalized capital law, skill premium.	14
2017	Public infrastructure		Tax-financed productive public input, opposing comparative-static effects.	15
2018	Three-period OLG		Age-indexed lifetime budget, life-cycle savings, aggregate saving.	16
2019	Old-age labor income		General saving rule with $e_2 > 0$, borrowing, dynamic efficiency.	7.6, 12
2020	Social Security		PAYG pension formula, log saving, capital law, tax comparative static.	13
2021	Participation		Indirect utility comparison, threshold, aggregate labor.	14
2022	Home production		Time allocation, type heterogeneity, market labor denominator.	14
2023	Endowment OLG		Loan clearing, no-trade condition, aggregate old/young resources.	7
2024	Non-tradable good	home	Intratemporal FOC, BGP with hours, home-productivity transition.	14, 11

23 Final Study Checklist

Before attempting old OLG quals, verify that you can do the following without notes.

1. Write a lifetime budget from sequential young/old constraints.

2. Derive $s = \frac{\beta}{1+\beta}e_1 - \frac{1}{1+\beta}\frac{e_2}{R}$.
3. Define a sequential competitive equilibrium in exchange OLG, government OLG, debt OLG, and production OLG.
4. Explain why loan-market clearing and goods feasibility are equivalent in the baseline exchange model.
5. Derive $(1+n)(1+g)k' = \frac{\beta}{1+\beta}(1-\alpha)k^\alpha$.
6. Solve the Cobb-Douglas log steady state and compute R^* .
7. State the difference between steady state in levels and BGP in normalized variables.
8. Compare R^* with $(1+n)(1+g)$ for dynamic efficiency.
9. Derive PAYG benefits from the government budget before solving the household problem.
10. Aggregate type-specific savings before normalizing.
11. Identify whether a shock changes capital, labor, effective labor, or prices on impact.
12. Evaluate Economist X claims by naming the exact object being compared.

Quality Audit

This companion is self-contained and includes source inventory, blueprint, module map, canonical derivations, equilibrium definitions, graphical intuition, closed-form catalog, traps, exercises, answer sketches, past-qual map, and checklist. The main limitation is that OG1.pdf and OG2.pdf are image-only PDFs in the uploaded files; their content was read from rendered page images rather than extracted text. The companion therefore avoids page-perfect quotation and instead reconstructs the models, definitions, and derivations in exam-ready notation.